Détection multi-label d'émotions en darija et en arabe algérien

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Code: https://github.com/Zorbiks/multilabel-emotions-darija

Introduction

A fine-tuned MARBERTv2 is the best model for ary (test macro-F1 0.553), and its joint ary+arq variant is the best for arq (0.552). We study multi-label detection of six emotions (joy, sadness, fear, anger, surprise, disgust) in Moroccan Darija (ary) and Algerian Arabic (arq). We compare trivial and TF-IDF baselines, zero-shot transfer, neural models trained from scratch, frozen and fine-tuned pre-trained encoders, and a multi-task presence+intensity model. All decisions (thresholds, model choice, early stopping) are taken on the validation set (= official dev); the test set is scored only once, at the end. We answer the twelve questions of the assignment and discuss the main limits: a small and noisy validation set, a strongly multi-label arq corpus, and very poor transfer between the two dialects.

1. Data and protocol

Splits are the official ones: ary 1608 / 267 / 812 and arq 901 / 100 / 902 (train / val / test); one duplicated arq training text is removed. Anything learned (vocabulary, TF-IDF, fastText, scaler) is fit on train only. The metric is the macro-F1 over the six emotions, and the decision threshold is tuned per emotion on val on a grid from 0.05 to 0.95. Neural models use three seeds (42, 43, 44).

Share of training texts (%)	joy	sadness	fear	anger	surprise	disgust	none
ary (n = 1608)	18.8	15.3	7.4	23.6	15.3	4.1	28.2
arq (n = 900)	17.0	44.8	24.8	32.9	34.8	22.9	10.1

The two corpora are very different. ary is mostly single-label (60.7% of texts have one emotion, 28.2% none), 17% of its texts are in Latin script (arabizi), and it contains news headlines and tweets. arq is almost entirely in Arabic script, narrative-like (47.9% of texts start with the narrator marker "al-rāwī :"), and strongly multi-label (about 58% of texts carry two or more emotions).

2. Answers to the questions

Q1 — Which metric, and what do trivial classifiers give?

We used macro-F1 over the 6 emotions. The labels are imbalanced and a text can have several emotions, so accuracy would mostly reflect the frequent labels. Macro-F1 gives every emotion the same weight.

Trivial classifiers on val:

Macro-F1	ary	arq
Predict "no emotion"	0.000	0.000
Most frequent emotion only	0.064	0.095
Predict all 6 emotions	0.240	0.435

On arq, "predict all" already gives 0.435, so our TF-IDF baseline (0.592) is not that far above it.

Q2 — Do surface cues explain the labels?

No, not really. The correlations with the labels are small. The strongest are emoji–joy 0.22 in ary, and "!?" with surprise 0.32 and anger 0.30 in arq. A simple emoji rule (laughing → joy, crying → sadness) gives macro-F1 0.008 on ary and 0.000 on arq, because these emoji are in only 0.4% of the val texts or absent. When we removed the suspected shortcut tokens, TF-IDF changed by at most 0.007.

Q3 — Does preprocessing matter? What about arabizi?

Almost no. All variants stay close to the default (0.4492 on ary, 0.5918 on arq). The only clear drop is when we delete the Latin/arabizi tokens: ary goes from 0.4492 to 0.4230. So arabizi carries useful signal.

Removing emoji and laughter at test time barely changes the scores (TF-IDF 0.4419 → 0.4410, MARBERTv2 0.5599 → 0.5529). Arabizi texts are harder: on ary test, MARBERTv2 gets macro-F1 0.172 on Latin-script texts and 0.574 on Arabic-script texts.

Q4 — How sensitive is zero-shot NLI to the wording?

Val macro-F1 (tuned thresholds)	en	fr	ar	ensemble
ary	0.341	0.344	0.344	0.356

Val macro-F1 (tuned thresholds)	en	fr	ar	ensemble
arq	0.508	0.495	0.482	0.498

With a fixed 0.5 threshold the wording matters more (arq: en 0.479 vs. ar 0.429). After tuning the thresholds the differences are small (at most 0.026). The ensemble helps on ary only. Zero-shot is below TF-IDF on both dialects (0.449 / 0.592).

Q5 — Sentiment model + rules: results and contamination

The rules (positive → joy, negative → anger + sadness) give macro-F1 0.214 (ary) and 0.230 (arq) with hard rules, and 0.236 / 0.249 with soft scores. Fear, surprise and disgust get F1 = 0 because the rules never predict them. For ary, sadness has high recall (0.78) but low precision (0.24), and joy has precision 0.66 but recall 0.42.

Contamination: we did not check it. The notebook only says to read the model card to see if its training data could overlap with BRIGHTER, and we did not do an overlap test, so we cannot say.

Q6 — Class imbalance: weights, oversampling, thresholds

The threshold has the biggest effect. For TF-IDF without class weights, fixed 0.5 → single threshold → per-label thresholds gives 0.157 → 0.379 → 0.419 on ary and 0.373 → 0.538 → 0.580 on arq. With balanced weights: 0.332 → 0.425 → 0.449 (ary) and 0.497 → 0.552 → 0.592 (arq).

For the char-CNN with per-label thresholds, plain loss, positive weights and oversampling are close: ary 0.437 / 0.453 / 0.453 and arq 0.546 / 0.549 / 0.538.

The weak labels are disgust (66 training positives in ary, F1 0.235 with TF-IDF) and fear (F1 0.424). In arq the weakest are disgust (0.512) and joy (0.545).

Q7 — Models from scratch: embeddings, attention, positional encoding

Val macro-F1 (mean ary/arq)	Score
C1 BiLSTM, random embeddings	0.434
C1 BiLSTM, fastText	0.449
C1 BiLSTM, AraVec	0.480
C2 BiLSTM + attention	0.482
C3 Transformer	0.481

Val macro-F1 (mean ary/arq)	Score
C3 without positional encoding	0.486
C4 char-CNN	0.486
TF-IDF (reference)	0.521

AraVec embeddings are the best of the three. Attention (C2) and the positional encoding did not help: C2 is about the same as C1, and the Transformer without positions is slightly better than with them. All models from scratch stay below TF-IDF.

For the attention, content words get more attention than their share of tokens (lift 1.06 in ary and 1.18 in arq), while punctuation (0.46 and 0.15) and emoji (0.50 in ary) get less.

> **[FIGURE PLACEHOLDER — attention weights]** Take a screenshot of the attention examples in notebook cell 36 (C2, ary/arq) and/or cell 40 (D2, 5 examples per dialect), and insert it here.

Q8 — Frozen vs. fine-tuned, DarijaBERT vs. MARBERTv2

Model	Val F1 ary	Val F1 arq	Train time ary / arq (s)	Trainable params
D1 frozen, [CLS] + LR	0.455	0.538	4.6 / 5.4	4.6 k
D2 frozen + BiLSTM-attention	0.518	0.572	8.4 / 5.0	0.95 M
E1 DarijaBERT fine-tuned	0.538	0.558	30.6 / 16.8	147 M
E1 MARBERTv2 fine-tuned	0.568	0.567	32.8 / 18.1	163 M

Fine-tuning helps on ary, but on arq the frozen D2 model (0.572) is as good as the fine-tuned ones and is cheaper to train. MARBERTv2 is better than DarijaBERT on both dialects (0.568 vs. 0.538 on ary, 0.567 vs. 0.558 on arq). In the final table, TF-IDF takes 2.5 ms per text on CPU and the transformer models about 64 ms.

Q9 — Presence + intensity (λ) and joint training

arq	val macro-F1	val r	test r
λ = 0	0.567	—	—
λ = 0.1	0.560	0.091	0.069

arq	val macro-F1	val r	test r
λ = 0.3	0.564	0.182	0.144
λ = 1	0.543	0.300	0.253
λ = 3	0.502	0.380	0.316

When λ goes up, the intensity correlation r goes up but the presence F1 goes down. We kept λ = 3 for Track B because it has the best val r, and r was still increasing, so a bigger λ could be tried.

With joint ary+arq training, arq improves on val (0.567 → 0.610) and ary gets slightly worse (0.568 → 0.559).

Q10 — Prior shift between val and test

In ary the label frequencies hardly move from val to test (at most +1.3 points, for surprise). In arq they move more: sadness +4.9, joy +4.7, disgust −5.6 points. Using the val thresholds instead of the best test thresholds costs 0.020 macro-F1 on ary (0.553 vs. 0.572, mostly fear) and 0.013 on arq (0.552 vs. 0.564).

In production, we would monitor the predicted-positive rate per emotion against the expected frequency, re-tune or recalibrate the thresholds (Platt, isotonic or prior-shift correction) on a small recent labelled sample, and follow F1 on a rolling audit set.

Q11 — Unseen dialect

MARBERTv2, test macro-F1	In-dialect	Transfer
train arq → test ary	0.550	0.213
train ary → test arq	0.469	0.214

Transfer is very bad: the score drops by more than half in both directions. The biggest drops per emotion are joy from arq to ary (0.653 → 0.047) and disgust from ary to arq (0.405 → 0.006).

Q12 — Error analysis and final comparison

We labelled 30 errors by hand (15 per dialect): multi-label 50.0%, implicit emotion 16.7%, dialect/code-switching 13.3%, irony/sarcasm 10.0%, other 6.7%, annotation noise 3.3% (1 in ary, 0 in arq). So most errors are multi-label mistakes.

Final test results (model chosen on val):

Dialect	Model	Test macro-F1	95% CI
ary	MARBERTv2 fine-tuned	0.553	[0.513, 0.584]
arq	MARBERTv2 joint	0.552	[0.532, 0.569]

The McNemar test between the two best models gives p = 0.0004 (ary) and p = 0.0005 (arq). On arq, TF-IDF reaches 0.504 test macro-F1 with only 2.5 ms per text.

3. Additional analysis: where the models fail

The confusion matrices below show, for the retained models, the per-emotion 2×2 confusion matrices and the matrix P(predicted j | true i), which reveals which emotions are confused with each other. All four figures come from notebook cell 69 (Part G); the PNG files can be extracted from its outputs (right-click → save image).

> **[FIGURE 1 — ary, MARBERTv2: per-emotion 2×2 confusion matrices on test]**
> Notebook cell 69, 1st output.

> **[FIGURE 2 — ary: P(predicted j | true i)]** Notebook cell 69, 2nd output.

> **[FIGURE 3 — arq, MARBERTv2 joint: per-emotion 2×2 confusion matrices on test]**
> Notebook cell 69, 3rd output.

> **[FIGURE 4 — arq: P(predicted j | true i)]** Notebook cell 69, 4th output.

Reading guide: [Add 2–3 sentences after looking at the figures: which emotions are most over-predicted, and which pairs (for instance anger/disgust or sadness/anger) are confused.]

4. Limitations and conclusion

- The arq validation set has only 100 texts, so thresholds, model selection and the val → test drop are noisy.
- Neural results use three seeds; differences below about 0.01–0.02 should not be interpreted.
- Contamination of the pre-trained models (Q5) and the causes of poor dialect transfer (Q11) were not tested.

- The error analysis is based on 30 examples and a single annotator.

Conclusion. Fine-tuned MARBERTv2 gives the best test macro-F1 (about 0.55) on both dialects, but a TF-IDF baseline is competitive on arq. Per-label threshold tuning on val is the single most effective trick for this task, surface cues and shortcuts play a minor role, the intensity objective competes with presence prediction, and dialect transfer fails. Future work: an ensemble of seeds with calibration, larger validation data (cross-validation), and explicit multi-label (label-correlation) modelling.